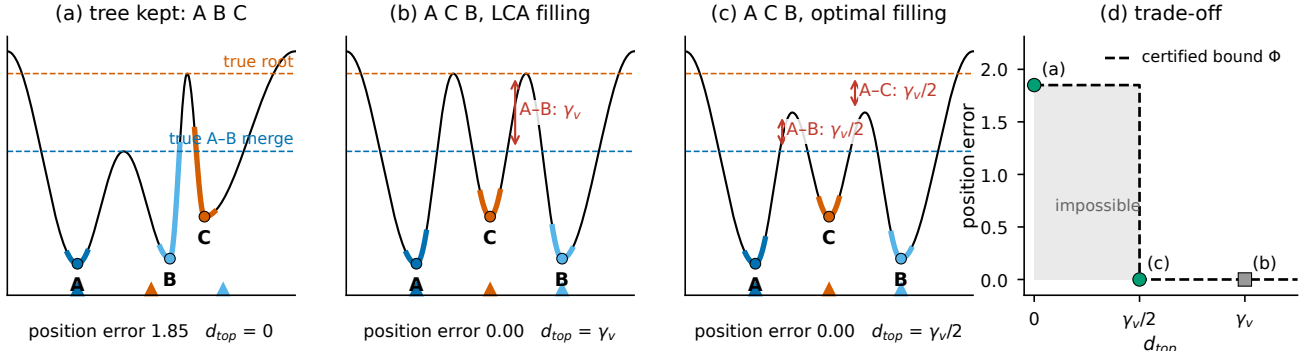
(a) tree kept: A B C
true root
true A–B merge
A
B
C
position error 1.85 $d_{top} = 0$
(b) A C B, LCA filling
A–B: γ_v
A
C
B
position error 0.00 $d_{top} = \gamma_v$
(c) A C B, optimal filling
A–C: $\gamma_v/2$
A–B: $\gamma_v/2$
A
C
B
position error 0.00 $d_{top} = \gamma_v/2$
(d) trade-off
certified bound Φ
(a)
impossible
(c)
(b)
position error
2.0
1.5
1.0
0.5
0.0
0
$\gamma_v/2$
γ_v
d_{top}